Groups, rings and modules

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Lecture 1 – 22/01/26

§1 Groups

§1.1 Groups – a reminder

Definition 1.1. A *group* is a triple $(G, \cdot, e)$ where G is a set,

$$\cdot : G \times G \to G$$

is a function, and $e \in G$ such that

- *Associativity.* $(a \cdot b) \cdot c = a \cdot (b \cdot c)$;
- *Identity.* For all $a \in G$, $a \cdot e = e \cdot a = a$;

- *Inverses*. For all $a \in G$, there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

The size of G, $|G|$, is called the *order* of the group.

Proposition 1.2 (Uniqueness of inverses)
Inverses are unique.

Proof. Left as an exercise (see 1A). ■

Definition 1.3 (Subgroups)**.** If G is a group and $H \subseteq G$, then we say H is a *subgroup* if

- $e \in H$;
- for all a, $b \in H$, $a \cdot b \in H$;
- $(H, \cdot, e)$ is a group.

In order to say item 3, we must first establish 1 and 2.

Proposition 1.4
A non-empty subset $H \subseteq G$ is a subgroup if and only if, for all h_1, $h_2 \in H$, $h_1 h_2^{-1} \in H$.

Proof. Left as an exercise (see 1A). ■

Definition 1.5. A group G is called *abelian* if, for all g_1, $g_2 \in G$, $g_1 g_2 = g_2 g_1$.

Example 1.6
- We have $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}/n\mathbb{Z}$.
- The symmetric groups S_n: permutations of $\{1, \dots, n\}$ under composition. More generally for any set X, we have $\mathrm{Sym}(X)$.
- The dihedral groups D_{2n}: the group of symmetries of a regular n-gon.
- The general linear group over $\mathbb{R}$, $\mathrm{GL}(n, \mathbb{R})$ or $\mathrm{GL}_n(\mathbb{R})$, is the set of $\{f : \mathbb{R}^n \to \mathbb{R}^n \mid f \text{ is an } \mathbb{R}\text{-linear bijection}\}$. Since $\mathrm{GL}(n, \mathbb{R})$ is made of functions, we can think of it a subset of $\mathrm{Sym}(\mathbb{R}^n)$. (Equivalently, it's the set of n by n invertible matrices with elements in $\mathbb{R}$).

Some examples of subgroups:

- The even permutations $A_n \leq S_n$ (the alternating group);
- Rotational symmetries of an n-goin, $C_n \leq D_{2n}$;
- Determinant one matrices $\mathrm{SL}(n, \mathbb{R}) \leq \mathrm{GL}(n, \mathbb{R})$.

Other examples include $C_2 \times C_2$, $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.

Definition 1.7. Let G be a group, $g \in G$ and $H \leq G$ a subgroup. Then the *left coset*

$$gH = \{gh : h \in H\}$$

Basic properties include that $g \in gH$, so cosets are non-empty. Also, cosets partition G – they have no non-trivial overlaps and every g lies in some coset. Furthermore, the map

$$\begin{aligned} H &\to gH \\ h &\mapsto gh \end{aligned}$$

defines a bijection.

Theorem 1.8 (Lagrange)
If G is a finite group and $H \leq G$ then

$$|G| = |H| \cdot |G : H|$$

where $|G : H|$, the index of H in G, is the number of cosets of H in $|G|$.

Definition 1.9. Let G be a group and let $g \in G$. The *order* of g, $\operatorname{ord}(g)$, is the smallest positive integer n such that $g^n = e$, or infinity if no such n exists.

Proposition 1.10
Let $g \in G$. Then, $\operatorname{ord}(g) \mid |G|$.

Proof. Let $g \in G$ and consider $H = \{e, g, g^2, \ldots, g^{n-1}\}$ where $n = \operatorname{ord}(g)$. This is a subgroup: it is non-empty, and $g^r \cdot g^{-s} = g^{r-s} \in H$ (after potentially multiplying by $g^n = e$). Also, the elements of H are distinct – if $g^i = g^j$ for $0 \leq i < j \leq n$, then $g^{j-i} = e$, contradicting the minimality of $\operatorname{ord}(g)$. Hence $|H| = ord(g)$ – now apply Lagrange. ∎

§1.2 Normal subgroups, quotients and homomorphisms

Let $H \leq G$. When are two cosets of H in G equal? If $gH = g'H$, then $g \in g'H$, so $g = g'h$ for some $h \in H$. In other words, $gH = g'H \iff (g')^{-1}g \in H$.

We denote the set of left cosets of $H \leq G$ by $G/H = \{gH : g \in G\}$. Does G/H carry a natural group structure?

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We propose the 'natural' group structure:

$$g_1H \cdot g_2H = (g_1g_2)H$$

But this rule may depend on the coset representatives we pick. We must check that it doesn't. Suppose $g_2H = g_2'H$, so $g_2' = g_2h$ for some $h \in H$. Then

$$g_1H \cdot g_2'H = (g_1g_2')H = g_1g_2hH = g_1g_2H$$

On the other hand, if $g_1H = g_1'H$, so $g_1' = g_1h$ for $h \in H$. Then

$$g_1'H \cdot g_2H = g_1hg_2H$$

and that is not automatically equal to g_1g_2H – we need $g_2^{-1}hg_2 \in H$ to hold for all $g_2 \in G$, $h \in H$.

Definition 1.11 (Normal subgroups)**.** A subgroup $H \leq G$ is *normal* if, for all $g \in G$, $h \in H$, $ghg^{-1} \in H$. We write this as $H \lhd G$.

> **Remark 1.12.** When $H \lhd G$, then G/H inherits a group structure by the rule above. We will refer to this as the *quotient group of G by H*.

Definition 1.13 (Homomorphisms)**.** Let G and H be groups. A homomorphism from G to H is a function $f : G \to H$ such that, for all g_1, $g_2 \in G$, $f(g_1g_2) = f(g_1)f(g_2)$.

Results such as $f(e_G) = e_H$ follow immediately.

Lemma 1.14

If $f : G \to H$ is a homomorphism, then $f(g^{-1}) = f(g)^{-1}$.

Proof. Calculate $f(g \cdot g^{-1})$ in two ways.

1. $f(gg^{-1}) = f(e_G) = e_H$;
2. $f(gg^{-1}) = f(g)f(g^{-1})$

so $f(g)f(g^{-1}) = e_H$ and the result follows. ∎

Definition 1.15. Let $f : G \to H$ be a homomorphism. The *kernel* of f is

$$\ker f = \{g \in G : f(g) = e_H\}$$

and the *image* of f is

$$\operatorname{im} f = \{h \in H : h = f(g) \text{ for some } g\}$$

Proposition 1.16

Let $f : G \to H$ be a homomorphism. Then $\ker f \lhd G$ and $\operatorname{im} f \leq H$.

Proof. Observe $e \in \ker f$ so $\ker f$ non-empty. Now, if $x, y \in \ker f$, then

$$f(xy^{-1}) = f(x)f(y)^{-1} = e$$

so $xy^{-1} \in \ker f$. For normality, let $x \in G$ and $g \in \ker f$. Now

$$f(xgx^{-1}) = f(x)ef(x)^{-1} = e$$

so $xgx^{-1} \in \ker f$ and hence $\ker f \lhd G$.

For the image, observe that, if a, $b \in \operatorname{im} f$ with $a = f(x)$, $b = f(y)$, then $ab^{-1} = f(xy^{-1}) \in \operatorname{im} f$. ∎

> **Remark.** Observe $f : G \to G/H$ taking $g \mapsto gH$ is a homomorphism. This is a nice way of expressing its 'naturalness'.

Definition 1.17. A homomorphism $f : G \to H$ is an *isomorphism* if it is a bijection. Two groups G and H are said to be *isomorphic*, $G \cong H$, if there exists an isomorphism between them.

Theorem 1.18 (First isomorphism theorem)
Let $f : G \to H$ be a homomorphism, then $\ker f$ is normal in G and the homomorphism

$$\begin{aligned} G/\ker f &\to \operatorname{im} f \\ g\ker f &\mapsto f(g) \end{aligned}$$

is an isomorphism. In particular, $G/\ker f \cong \operatorname{im} f$.

Proof. We have already shown the normality result. Consider the map

$$\begin{aligned} \varphi : G/\ker f &\to \operatorname{im} f \\ g\ker f &\mapsto f(g) \end{aligned}$$

- φ *is well-defined*. If $g\ker f = g'\ker f$, then $g'g^{-1} \in \ker f$, so $f(g'g^{-1}) = e$ which gives $f(g) = f(g')$ as required.
- φ *is surjective*. Let $h \in \operatorname{im} f$. Then $h = f(g)$ for some $g \in G$, and so $h = \varphi(g\ker f)$.
- φ *is injective*. Suppose $\varphi(g\ker f) = \varphi(g'\ker f)$, then $f(g) = f(g')$. But then $g'g^{-1} \in \ker f$, so $g\ker f = g'\ker f$.

■

Example
Consider the homomorphism

$$\begin{aligned} \exp : (\mathbb{C}, +) &\to (\mathbb{C}^*, \times) \\ z &\mapsto e^z \end{aligned}$$

Then $\ker \exp = 2\pi i\mathbb{Z}$, so $\mathbb{C}^* \cong \mathbb{C}/2\pi i\mathbb{Z}$.

Theorem 1.19 (Second isomorphism theorem)
Let $H \leq K$ and $K \lhd G$. Then

$$HK = \{hk : h \in H, k \in K\}$$

is a subgroup of G. Furthermore, $H \cap K$ is normal in H, and there is an isomorphism $HK/K \cong H/H \cap K$.

Proof. We will show the results in order.

- *HK is a subgroup.* It contains e_G, so is non-empty. Now consider $x = hk$ and $y = h'k'$. We'll show $yx^{-1} \in HK$. Observe

$$\begin{aligned} yx^{-1} &= h'k'(hk)^{-1} \\ &= h'k'k^{-1}h^{-1} \\ &= (h'h^{-1})h(k'k^{-1})h^{-1} \\ &= h'h^{-1}k'' \end{aligned}$$

  by normality of K.

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- *$H \cap K$ is a kernel.* Consider

$$\begin{aligned} \varphi : H &\to G/K \\ h &\mapsto hK \end{aligned}$$

  by restricitng the domain from $G \to G/K$ – the kernel is $H \cap K$.

- *Final statement.* By the first isomorphism theorem, we have

$$\operatorname{im} \varphi \cong H/(H \cap K)$$

  so we must show $\operatorname{im} \varphi \cong HK/K$. But $\operatorname{im} \varphi$ is the set of K-cosets in G that are representable by elements of H, i.e. $\{hK : h \in H\} \subseteq G/K = HK/K$.

∎

Theorem 1.20 (Correspondence between subgroups of G and G/K)
Let $p : G \to G/K$ be the quotient homomorphism. We have a bijection

$$\begin{aligned} \{\text{subgroups of } G \text{ containing } K\} &\leftrightarrow \{\text{subgroups of } G/K\} \\ K \triangleleft L \leq G &\mapsto L/K \leq G/K \\ \{g \in G : gK \in A\} &\leftarrow\!\shortmid A \leq G/K \end{aligned}$$

Proof. The forward bijection is just applying the projection map on a subset, and, since the quotient map is a homomorphism, its image is a subgroup; and the backward bijection is actually just applying the preimage of p. It is an exercise to check that the forward and backward directions are inverses. ∎

> **Remark.** In fact the correspondence matches normal subgroups on the two sides.

Theorem 1.21 (Third isomorphism theorem)

Let $K \le L \le G$ be normal subgroups of G. Then there exists an isomorphism $(G/K)/(L/K) \cong G/L$.

Proof. Define a map

$$\begin{aligned} \varphi : G/K &\to G/L \\ gK &\mapsto gL \end{aligned}$$

- *φ is well-defined.* If $gK = g'K$, then $g'g^{-1} \in K \subseteq L \implies gL = g'L$.
- *φ is a homomorphism.* Clear.

We have $\ker\varphi = \{gK : gL = L\} = \{gK : g \in L\} = L/K$. Hence we are done by the first isomorphism theorem. ■

Definition 1.22 (Simplicity)**.** A group G is called *simple* if its only normal subgroups are G and $\{e\}$.

Proposition 1.23

Let G be an abelian group. Then G is simple if and only if $G \cong C_p$ for some prime p.

Proof. If $G \cong C_p$, then any non-identity element generates G by, for example, Lagrange. So G is simple.

If G is simple and abelian, then take any $g \neq e \in G$ and consider $\langle g \rangle$. This is a subgroup, and since G is abelian, it is automatically normal. Hence, by simplicity, $\langle g \rangle = G$, and so G is cyclic.

If G is infinite and cyclic, then $G \cong \mathbb{Z}$, but $\mathbb{Z}$ is not simple, so this is a contradiction. Hence G is finite and $G \cong C_m$. If q divides m, take the subgroup generated by $g^{m/q}$, where g generates G. Normality is automatic, so $q = 1$ or $q = m$ by simplicity – in other words, m is prime. ■

Theorem 1.24

Let G be a finite group. Then, there exist subgroups

$$G = H_1 \triangleright H_2 \triangleright H_3 \triangleright \cdots \triangleright H_n = \{e\}$$

such that H_i/H_{i+1} is simple for all i.

Proof. If G is simple, $H_2 = \{e\}$, and we're done. Otherwise, let H_2 be a proper normal subgroup of maximal order in G. We claim G/H_2 is simple. If not, consider the quotient map $\pi : G \to G/H_2$.

If $N \triangleleft G/H_2$ is a proper normal subgroup, by the correspondence theorem, we can find a proper normal subgroup in G containing H_2. But this contradicts the maximality of H_2.

Now we can just iterate – note that $|H_{i+1}| < |H_i|$, so we will eventually terminate. ■

> **Remark.** This sequence is not necessarily unique.

§1.3 Group actions and permutations

Definition 1.25. Let X be a set. Let $\mathrm{Sym}(X)$ be the set of permutations on X. The identity is the identity function, and the group law is given by composition. If $X = \{1, \ldots, n\}$, then we write $\mathrm{Sym}(X) = S_n$.

- It's convenient to write $\sigma \in S_n$ as a product of disjoint cycles.
- Every $\sigma \in S_n$ is a product of transpositions, and the number of such transpositions gives rise to the *sign* of σ.

  $$\begin{aligned} \mathrm{sign} : S_n &\to \{\pm 1\} \\ \sigma &\mapsto \begin{cases} +1 & \sigma \text{ is a product of an even number of transpositions} \\ -1 & \sigma \text{ ... odd ...} \end{cases} \end{aligned}$$

  sign is a homomorphism, and is surjective when $n \geq 3$.

Let X be a set and G a group, and consider a homomorphism $G \to \mathrm{Sym}(X)$. This is called a *permutation representation* of G.

Definition 1.26 (Group actions)**.** An *action* of a group G on a set X is a function

$$\begin{aligned} G \times X &\to X \\ (g, x) &\mapsto g \cdot x \end{aligned}$$

such that

- $e \cdot x = x$ for all $x \in X$;
- $g_1 \cdot (g_2 x) = (g_1 g_2) \cdot x$ for all $g_1, g_2 \in G$ and $x \in X$.

We say $G \curvearrowright X$.

We should note the bijection between permutation representations and group actions.

Proposition 1.27

The map a is a bijection.

$$\begin{aligned} a : \{\text{homomorphisms } G \to \mathrm{Sym}(X)\} &\longrightarrow \{\text{actions } G \curvearrowright X\} \\ \varphi : G \to \mathrm{Sym}(X) &\longmapsto a(\varphi) : G \times X \to X : (g, x) \mapsto \varphi(g) \cdot x \end{aligned}$$

Proof. Let us simply construct an inverse. Let $\cdot : G \times X \to X$ be an action. Define

$$\begin{aligned} \varphi : G &\to \mathrm{Sym}(X) \\ g &\mapsto \varphi(g) := [g \cdot : X \to X] \end{aligned}$$

This is a permutation as $(g^{-1})\cdot : X \to X$ is the inverse. Indeed,

$$\begin{aligned}\varphi(g^{-1})(\varphi(g)(x)) &= g^{-1} \cdot (g \cdot x) \\ &= (g^{-1}g)x \\ &= x\end{aligned}$$

To see that this rule gives a homomorphism is simple. ■

We have some notation:

- $G^X = \operatorname{im}\varphi \leq \operatorname{Sym}(X)$;
- $G_X = \ker\varphi \lhd G$

It is a special case of the first isomorphism theorem that $G^X \cong G/G_X$.

Example 1.28

Let G be the symmetry group of the unit cube, and let X be the set of the cube's diagonals. Clearly G acts on X. We have hence

$$\varphi : G \to \operatorname{Sym}(X) \cong S_4$$

We can see $\operatorname{im}\varphi \cong S_4$, since we can swap any two diagonals – then these transpositions generate the whole symmetric group. Furthermore, $\ker\varphi = \{e, s\}$, where s sends every vertex to its opposite. As an immediate consequence, we conclude $|G| = 48$.

Example 1.29

For any group G, we can make G act on itself (as a set) by

$$\begin{aligned} G \times G &\to G \\ (g, g_1) &\mapsto gg_1 \end{aligned}$$

which gives $\varphi : G \to \operatorname{Sym}(G)$. The kernel $\ker\varphi = \{e\}$, and so φ is injective – we conclude that G is isomorphic to a subgroup of $\operatorname{Sym}(G)$ by the first isomorphism theorem.

Example 1.30

Let $H \leq G$ and $X = G/H$. Then $G \curvearrowright X$ by

$$(g, g_1 H) \to g g_1 H$$

We get a homomorphism $\varphi : G \to \mathrm{Sym}(X)$. Consider the kernel – it is clear that $\ker \varphi \subseteq H$. But more is true. If $g \in \ker \varphi$, then

$$\begin{aligned} g g_1 H &= g_1 H \\ \implies g_1^{-1} g g_1 &\in H \\ \implies \ker \varphi &\subseteq \bigcap_{g_1 \in G} g_1 H g_1^{-1} \end{aligned}$$

that is, g is contained in every conjugate of H. Furthermore, this containment is reversible: suppose $g \in \bigcap_{g_1 \in G} g_1 H g_1^{-1}$. Then $g g_1 H = g_1 H$, so $g \in \ker \varphi$. We conclude

$$\ker \varphi = \bigcap_{g_1 \in G} g_1 H g_1^{-1}$$

If H were normal, we would just have H on the right-hand side – but when H is not normal, we have an interesting way to make a smaller normal subgroup, $\ker \varphi$, out of H.

Theorem 1.31

Let G be finite and let $H \leq G$ be of index n. Then, there exists a normal $K \triangleleft G$ with $K \subseteq H$ such that G/K is isomorphic to a subgroup of S_n. Thus $|G/K|$ divides $n!$, and $|G/K| \geq n$.

Proof. Consider $\varphi : G \to \mathrm{Sym}(G/H)$ from the example. The kernel, call it K, is normal. We have shown it is contained in H already – applying the first isomorphism theorem, we have

$$G/K \cong \mathrm{im}\, \varphi \leq \mathrm{Sym}(G/H) \cong S_n$$

Lagrange gives $|G/K|$ divides $n!$, and $K \leq H$, so $|G/K| \geq |G/H| = n$. ■

Corollary 1.32

Let G be a non-abelian simple group. Let $H \leq G$ be a proper subgroup with index $n > 1$. Then G is isomorphic to a subgroup of A_n, and we must have $n \geq 5$.

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Proof. The action of G on $X = G/H$ gives a homomorphism

$$\phi : G \to \mathrm{Sym}(X) \cong S_n$$

Since $\ker\phi$ is normal, have $\ker\phi = G$ or $\{e\}$ by simplicity. If $\ker\phi = G$, every element acts trivially, but, since H is a proper subgroup, this is impossible, since some $g \in G$ gives $gH \neq H$. Hence $\ker\phi = \{e\}$.

Thus, the first isomorphism theorem gives

$$G \cong \operatorname{im}\phi \leq \operatorname{Sym}(X) \cong S_n$$

and we would like to show $\operatorname{im}\phi \leq A_n$. Observe $A_n \triangleleft S_n$, and, as G is simple, $\operatorname{im}\phi \cap A_n$ is either $\{e\}$ or $\operatorname{im}\phi$ itself.

The second isomorphism theorem gives that, if $\operatorname{im}\phi \cap A_n = \{e\}$,

$$\operatorname{im}\phi \cong \frac{\operatorname{im}\phi}{\operatorname{im}\phi \cap A_n} \cong \frac{\operatorname{im}\phi \cdot A_n}{A_n} \leq S_n/A_n \cong C_2$$

which would mean $\operatorname{im}\phi$ is abelian, a contradiction.

It remains to check that $n \geq 5$. This follows from the fact that S_1, S_2 are abelian, and S_3, S_4 have no non-abelian simple subgroups. ■

Definition 1.33 (Orbit and stabiliser)**.** Let $G \curvearrowright X$ be a group action. The *orbit* of $x \in X$ is

$$Gx = \{gx \mid g \in G\} \subseteq X$$

and the *stabiliser* of $x \in X$ is

$$G_x = \{g \in G \mid gx = x\} \subseteq G$$

Theorem 1.34 (Orbit-stabiliser)

Suppose $G \curvearrowright X$, $x \in X$. Then there exists a bijection

$$\begin{aligned} Gx &\to G/G_x \\ gx &\mapsto gG_x \end{aligned}$$

In particular, if G is finite, have

$$|G| = |Gx||G_x|$$

for any $x \in X$.

Proof. Exercise. ■

§1.4 Conjugacy, centralisers and normalisers

A group G can act on itself by conjugation

$$(g, h) \mapsto ghg^{-1}$$

which of course gives a homomorphism $G \to \operatorname{Sym}(G)$. Now, fix $g \in G$ and take the following map:

$$\begin{aligned} G &\to G \\ h &\mapsto ghg^{-1} \end{aligned}$$

This is both a permutation and a homomorphism. This motivates the following definition.

Definition 1.35 (Automorphism)**.** Let G be a group. A permutation

$$a : G \to G$$

is called an *automorphism* if it is also a homomorphism.

> **Remark.** It is easy to see that the set of all automorphisms
>
> $$\mathrm{Aut}(G) = \{f : G \to G \mid f \text{ an automorphism}\} \subseteq \mathrm{Sym}(G)$$
>
> is in fact a subgroup of $\mathrm{Sym}(G)$.

Definition 1.36 (Conjugacy class, centraliser and centre)**.** Fix $g \in G$. The *conjugacy class* of g is the the orbit of g under conjugation, that is

$$\mathrm{ccl}_G(g) = \{hgh^{-1} \mid h \in G\}$$

The *centraliser* of g is the stabiliser of g, that is

$$C_G(g) = \{h \in G \mid hgh^{-1} = g\}$$

The *centre* of G is the kernel of the action homomorphism, that is

$$Z(G) = \bigcap_{g \in G} C_G(g) = \{z \in G \mid gzg^{-1} = z \text{ for any } g \in G\}$$

Proposition 1.37

Let G be a finite group. Then

$$|\,\mathrm{ccl}(x)| = |G \colon C_G(x)| = |G|/|C_G(x)|$$

Proof. Apply orbit-stabiliser to the conjugation action. ∎

Definition 1.38 (Normaliser)**.** Let $H \leq G$. The *normaliser* of H in G is given by

$$N_G(H) = \{g \in G \mid gHg^{-1} = H\}$$

Observe that $H \lhd N_G(H)$. By definition, this is the largest subgroup of G in which H is normal.

§1.5 Simplicity of the alternating groups A_n $(n \geq 5)$

Recall that, in S_n, a conjugacy class is a collection of elements all of the same cycle type.

Theorem 1.39

Let $n \geq 5$. Then A_n is simple.

When $n = 5$, it is simple to check which conjugacy classes in S_n split in A_n, and then use the fact that any normal subgroup is a union of conjugacy classes. For $n \geq 5$, we will need a more powerful strategy. We prove the following three claims:

1. A_n is generated by 3-cycles;
2. any $H \lhd A_n$ that contains a 3-cycle contains every 3-cycle;
3. any non-trivial $H \lhd A_n$ contains a 3-cycle.

Claim 1 – A_n is generated by 3-cycles. Observe that any $g \in A_n$ is a product of an even number of transpositions (when viewed in S_n). Consider a product of two transpositions – there are three possibilities:

$$(a\,b)(a\,b) = e$$
$$(a\,b)(b\,c) = (a\,b\,c)$$
$$(a\,b)(c\,d) = (a\,c\,b)(a\,c\,d)$$

In each case, the product is a product of 3-cycles. The result follows.

Claim 2 – Any two 3-cycles are conjugate in A_n. Let δ, δ' be 3-cycles, and write

$$\delta' = \sigma\delta\sigma^{-1}$$

where $\sigma \in S_n$. If σ even, we're done, so suppose σ odd. Now, since $n \geq 5$, there exists a transposition τ that commutes with δ. Thus,

$$\delta' = \sigma\tau\delta\tau^{-1}\sigma^{-1} = (\sigma\tau)\delta(\sigma\tau)^{-1}$$

Since $\sigma\tau$ is even, we're done.

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Claim 3 – Any non-trivial $H \lhd A_n$ contains a 3-cycle.

- If H contains an element of the form $\sigma = (1\ 2\ \ldots\ \gamma)\tau$ with τ disjoint from $(1\ 2\ \ldots\ \gamma)$ and $\gamma \geq 4$, then normality of H gives that $\delta^{-1}\sigma\delta \in H$, so furthermore $\sigma^{-1}\delta^{-1}\sigma\delta \in H$. Since τ misses 1, 2, 3, it commutes with δ, and with $(1\ 2\ \ldots\ \gamma)$ by assumption.

  Then, expanding $\sigma^{-1}\delta^{-1}\sigma\delta$ out, we get $(2\ 3\ \gamma)$, so we have our 3-cycle.

- If H contains an element of the form $\sigma = (1\ 2\ 3)(4\ 5\ 6)\tau$ in a disjoint cycle decomposition, take $\delta = (1\ 2\ 4)$ and calculate $\sigma^{-1}\delta^{-1}\sigma\delta$ – this is a 5-cycle, bringing us into case 1.

- If H contains $\sigma = (1\ 2\ 3)\tau$ where τ is a product of disjoint transpositions (if τ contained anything longer, we could relabel and apply cases 1 or 2). Then σ^2 is a 3-cycle.

- If H contains $\sigma = (1\ 2)(3\ 4)\tau$ where τ is a product of disjoint transpositions, let $\delta = (1\ 2\ 3)$ and consider $\mu = \sigma^{-1}\delta^{-1}\sigma\delta = (1\ 4)(2\ 3)$. Then let $\nu = (1\ 5\ 2)\mu(1\ 2\ 5) = (1\ 3)(4\ 5)$ – since $\mu\nu = (1\ 2\ 3\ 4\ 5)$, we are in case 1.

This covers all cases up to relabelling.

§1.6 Finite p-groups

Let p be a prime. Groups of order p^n for some n are called *p-groups*.

Theorem 1.40

Let G be a p-group. Then the centre of G is non-trivial.

Proof. Let G act on itself by conjugation. The centre $Z(G)$ is the set of all size one orbits. Since the orbits partition G, we have

$$|G| = p^n = |Z(G)| + \sum \text{sizes of conjugacy classes of size} > 1$$

We know conjugacy classes have sizes dividing p^n. Hence, if a conjugacy class has size larger than 1, it is a multiple of p. Then $Z(G)$ is a difference of multiples of p and the result follows ■

Theorem 1.41

A group of size p^2 is abelian.

Lemma 1.42

If $G/Z(G)$ is cyclic, then G is abelian.

Proof of lemma. We have a coset $xZ(G)$ in $G/Z(G)$ that generates $G/Z(G)$, so every coset has the form $x^m Z(G)$. This means every G can be written as $g = x^m z$ for some m and $z \in Z(G)$. From this the result immediately follows. ■

Proof of theorem. By theorem 1.40, $Z(G)$ is either G or has size p. In the former case we are done – in the latter case, $G/Z(G) \cong C_p$, which is cyclic, so, by lemma 1.42, G is abelian. ■

Theorem 1.43

Let G be a p-group of size p^n. Then G has a subgroup of size k for all $0 \leq k \leq n$.

Proof. We induct on n. *Base case.* $n = 1$, the result is clear.s *Inductive step.* Suppose $n > 1$. If $k = 0$, take $\{e\}$. Otherwise, note $Z(G)$ is non-trivial. Let $x \in Z(G)$ with $x \neq e$, and its order is a power of p – if we replace x with some power of x, say x', we can produce an element of order p. Then $\langle x \rangle'$ has order p and is normal.

Now consider $G/\langle x \rangle'$. This is a group of size p^{n-1}, and so it contains a subgroup, say L. of size p^{k-1} by induction. By the correspondence of subgroups, we find $K \leq G$ containing $\langle x \rangle'$ such that $K/\langle x \rangle' \cong L$. But then $|K| = p^k$, as required. ■

§1.7 Finite abelian groups

Theorem 1.44 (Classification of finite abelian groups)
Let G be a finite abelian group. There exist positive integers $d_1, \dots, d_r$ such that

$$G \cong C_{d_1} \times \cdots \times C_{d_r}$$

Furthermore, we can choose the d_i such that $d_{i+1} \mid d_i$, in which case the expression is unique.

Example 1.45
Consider, for example, abelian groups of order 8 – they are exactly C_8, $C_4 \times C_2$, and $C_2 \times C_2 \times C_2$.

Lemma 1.46 (Chinese remainder theorem)
If n and m are coprime, then $C_n \times C_m \cong C_{nm}$.

Proof. It suffices to find an order mn element of $C_n \times C_m$. If $g \in C_n$ has order n and $h \in C_m$ has order m, then consider $(g, h) \in C_n \times C_m$. Suppose it has order k, then $g^k = h^k = e$, so that $n \mid k$ and $m \mid k$. But then $nm \mid k$, and $k \mid nm$ by Lagrange, so $k = mn$ as required. ∎

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Corollary 1.47
Let G be finite and abelian, then we have

$$G \cong C_{l_1} \times \cdots \times C_{l_n}$$

where each l_i is a prime power.

Proof. The result follows from theorem 1.44 and lemma 1.46. ∎

§1.8 Sylow theorems

Let G be a finite group of size $p^a m$ where $p \nmid m$ and p is some prime. A *Sylow subgroup* of G is a subgroup of order p^a.

Theorem 1.48 (Sylow theorems)

Let G be a finite group and p a prime.

1. The set $\mathrm{Syl}_p(G) = \{P \leq G : P \text{ is Sylow-}p\}$ is non-empty;
2. any $H,\, H' \in \mathrm{Syl}_p(G)$ are conjugate;
3. if $n_p = |\,\mathrm{Syl}_p(G)|$, then $n_p \equiv 1 \pmod p$, and $n_p \mid |G|$ (so $n_p \mid m$).

Corollary 1.49

$\mathrm{Syl}_p(G) = \{P\}$ if and only if P is normal.

Proof. For any $g \in G$, gPg^{-1} is isomorphic as a group to P, so it is also a Sylow p-subgroup. ■

Corollary 1.50

Let G be non-abelian and simple and suppose $p \mid |G|$. Then $|G|$ divides $\frac{n_p!}{2}$ and $n_p \geq 5$.

Proof. The conjugation action of G on $\mathrm{Syl}_p(G)$ gives a homomorphism

$$\varphi : G \to \mathrm{Sym}(\mathrm{Syl}_p(G)) \cong S_{n_p}$$

By simplicity of G, $\ker\varphi = \{e\}$ or G. If $\ker\varphi = G$, then $gPg^{-1} = P$ for all $g \in G$, so P is normal. Since $P \neq \{e\}$, $P = G$. But $Z(P) \triangleleft P$ non-trivially, which is a contradiction. Hence φ is injective, so G isomorphic to a subgroup of S_{n_p}. Consider now

$$G \xrightarrow{\varphi} S_{n_p} \xrightarrow{\text{sign}} \{\pm 1\}$$

If the composition is surjective, then $\ker(\text{sign} \circ \varphi)$ is an index 2 subgroup of G. But G is simple, so this is impossible, (since $G \neq C_2$). In other words, $\operatorname{im}\varphi \subseteq A_{n_p}$, and the result follows by Lagrange. Finally, all non-abelian subgroups of A_2, A_3 and A_4 are not simple. ■

Example

Suppose G has size $11 \times 12 = 132$. If G is simple, then none of the n_p's are equal to 1. Furthermore, there are 12 Sylow-11 subgroups, since $n_11 \equiv 1 \pmod 1 1$ and $n_11 \mid 12$.

Similarly, $n_3 \equiv 1 \pmod 3$, and $n_3 \mid 44$, so $n_3 = 4$ or 22 – by corollary 1.50, $n_3 = 22$. Any two Sylow 11-subgroups intersect only at $\{e\}$, so we already have 120 elements. We also have at least 44 elements of order 3, which is impossible, so no such simple G exists.

Proof of Sylow part 1. We will try and produce a Sylow p-subgroup as a stabiliser of an element in an appropriate action.

Let $|G| = n = p^a m$, of course $p \nmid m$. Let $\Omega = \{X \subseteq G : |X| = p^a\}$. Let G act on Ω by $g \cdot \{g_1, \dots, g_{p^a}\} = \{gg_1, \dots, gg_{p^a}\}$.

Claim 1. We have $|\Omega| = \binom{n}{p^a} = m \not\equiv 0 \pmod p$.

Proof. Expand $(1+x)^n$ mod p.

Claim 2. Suppose $U \in \Omega$ and $H \leq G$ stabilises U, then $|H| \mid |U|$.

Proof. We have $hU = U$ for all $h \in H$. In other words, for all $u \in U$, the coset Hu is contained in U. Furthermore, every $u \in U$ lies in some coset of H, and the cosets are disjoint, so the cosets partition U. We get $|H| \mid |U|$.

We now know $|\Omega| \neq 0 \pmod p$, and $|\Omega| = |O_1| + \cdots + |O_r|$ where the O_i are the orbits of the action. Since $p \nmid |\Omega|$, there exists an orbit, say Θ, whose size is coprime to p.

Let $T \in \Theta$. By the orbit-stabiliser theorem, $|G| = |\Theta| \cdot |\operatorname{Stab}(T)|$, so $p^a m = |\Theta||\operatorname{Stab}(T)|$. By claim 2, $|\operatorname{Stab}(T)|$ divides p^a, but there are no factors of p in Θ, so we are done. ■

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Proof of Sylow part 2. We will show a stronger statement, namely that if $Q \leq G$ is a p-group in G, and $P \in \operatorname{Syl}_p(G)$, there exists $g \in G$ s.t. $gQg^{-1} \subseteq P$. This clearly implies the claim.

Let Q act on G/P by $q \cdot gP = qgP$. The orbits of this action have size either 1 or divisible by p, but $|G/P|$ is not divisible by p, so there must be a size 1 orbit. In other words, have $g \in G$ such that $qgP = gP$ for all $q \in Q$. Rearranging, we obtain $g^{-1}qg \in P$ for all $q \in Q$, so $g^{-1}Qg \subseteq P$. ■

Proof of Sylow part 3. Recall $n_p = |\operatorname{Syl}_p(G)|$, and we would like to show $n_p \mid |G|$ and $n_p \equiv 1 \pmod p$. The first part is not so bad – the conjugation action of G on $\operatorname{Syl}_p(G)$ has one orbit of size n_p, whence the result.

Let $P \in \operatorname{Syl}_p(G)$, and consider P acting on $\operatorname{Syl}_p(G)$ by conjugation. Orbit-stabiliser gives that the orbits have size 1 or a multiple of p – but $\{P\}$ clearly constitutes a size 1 orbit. It suffices, then, to show that all other orbits have size divisible by p.

Say Q is another size 1 orbit, so, for every $h \in P$, we have $hQh^{-1} = Q$. Thus, P is contained in the normaliser $N_G(Q)$ of Q. Observe that

$$Q \leq N_G(Q) \leq G$$

so $Q \in \operatorname{Syl}_p(N_G(Q))$. But, by definition, $Q \lhd N_G(Q)$, and P is another Sylow p-subgroup of $N_G(Q)$, so therefore $Q = P$. Hence, there is only one orbit of size 1, so the result follows. ■

Example

Let $G = \mathrm{GL}_n(\mathbb{Z}/p\mathbb{Z})$, the set of invertible n by n matrices with integer entries mod p. What is $|G|$? A bad way – to start with p^{n^2}, and remove the singular matrices. Instead, let us consider building it column by column:

$$M = \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{pmatrix}$$

For $\mathbf{v}_1$, we can have any non-zero vector, so $p^n - 1$ choices. For $\mathbf{v}_2$, we need to pick something linearly independent from $\mathbf{v}_1$, i.e. something which is not a multiple of $\mathbf{v}_1$. There are $p^n - p$ such choices. Similarly, $\mathbf{v}_3$ should not be of the form $a\mathbf{v}_1 + b\mathbf{v}_2$, so there are $p^n - p^2$ choices – etc. Hence,

$$|G| = (p^n - 1)(p^n - p) \cdots (p^n - p^{n-1})$$

What is the largest power of p dividing this? – $p^{\binom{n}{2}}$. Let's look for a Sylow p-subgroup. Motivated by the fact that $\binom{n}{2}$ counts triangle numbers, we consider the subgroup of upper triangular matrices with 1 on the diagonal. Thanks to the second Sylow theorem, we've found all of the Sylow p-subgroups, since we can just conjugate!

§2 Rings

§2.1 Basic definitions

Definition 2.1 (Ring)**.** A *ring* is a quintuple $(R, +, \cdot, 0_R, 1_R)$, where R is a set, 0_R, $1_R \in R$, and

$$\begin{aligned} + &: R \times R \to R && \text{(addition)} \\ \cdot &: R \times R \to R && \text{(multiplication)} \end{aligned}$$

such that

- $(R, +, 0_R)$ is an abelian group;
- the operation $\cdot$ is associative, and 1_R is a multiplicative unit;
- multiplication is distributive over addition, that is $a \cdot (b + c) = a \cdot b + a \cdot c$, and $(b + c) \cdot a = b \cdot a + c \cdot a$.

As usual, we will write R for the ring. The symbol $-r$ will refer to the additive inverse of r, $r - s$ will refer to $r + (-s)$, etc.

> **Remark.** We will take R to be commutative, that is $ab = ba$ for all $a, b \in R$. The additive structure is always abelian.

Definition 2.2 (Subring)**.** A *subring* of a ring R is a subset $S \subseteq R$ such that 0_R, $1_R \in S$, S is closed under addition and multiplication, and $(S, +, \cdot, 0_R, 1_R)$ is a ring. We write $S \leq R$.

Example

We have the familiar number systems $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$. We also have the mildly less familiar $\mathbb{Z}[i] \leq \mathbb{C}$ (the *Gaussian integers*), and other fun stuff like $\mathbb{Q}[\sqrt{2}]$. Weirder examples include $\mathbb{C}\langle\varepsilon\rangle = \{a + b\varepsilon \mid a, b \in \mathbb{C}\}$ where

$$(a + b\varepsilon) + (c + d\varepsilon) = (a + c) + (b + d)\varepsilon$$
$$(a + b\varepsilon) \cdot (c + d\varepsilon) = ac + (ad + bc)\varepsilon$$

or in other words $\varepsilon^2 = 0$. These are called the *dual numbers*.

Definition 2.3 (Units)**.** An element $u \in R$ is an *unit* if there exists $v \in R$ such that $uv = 1$.

Remark. This notion does not interact well with subrings. For example, 2 is a unit in $\mathbb{Q}$, but not in $\mathbb{Z}$.

Definition 2.4 (Field)**.** A ring R is a *field* if every non-zero element is a unit.

Proposition 2.5

Let R be a ring with additive identity 0. Then, for any $r \in R$, $0 \cdot r = 0$.

Proof. Have

$$\begin{aligned} 0 + 0 &= 0 \\ \implies r \cdot 0 + r \cdot 0 &= r \cdot 0 \\ \implies r \cdot 0 &= 0 \end{aligned}$$

■

Remark. If $1 = 0$, we have $r = 1 \cdot r = 0 \cdot r = 0$, so we have the *zero ring*.

Definition 2.6 (Ring direct product)**.** Let R, S be rings. The *direct product* $R \times S$ is a ring with operations and identities defined pointwise.

Definition 2.7 (Polynomials)**.** Let R be a ring. A *polynomial* in X with coefficients in R is an expression of the form

$$f(X) = a_0 + a_1 X + \cdots + a_n X^n$$

where each $a_i \in \mathbb{R}$, and the X^i are formal symbols. As convention, we will identify $f(X)$ with $f(X) + 0 \cdot X^{n+1}$ – then, the largest n with $a_n \neq 0$ is the *degree* of $f(x)$ (by convention, we say the degree of 0 is $-\infty$). Finally, a polynomial is *monic* if it ha s degree n and $a_n = 1$.

Definition 2.8 (Polynomial rings)**.** Let R be a ring. The *polynomial ring* $R[X]$ is the set of all polynomials with coefficients in R, with obvious operations and identities.

> **Remark.** Note that, for every $f \in R[X]$, we get a function $R \to R$ which takes r to $f(r)$. But this is not the same as the polynomial itself.

Definition 2.9 (Rings of formal power series)**.** The ring of *formal power series* over R in the variable X is

$$R[[X]] = \left\{ \sum_{i=0}^{\infty} r_i X^i : r_i \in R \right\}$$

with obvious operations and identities.

Example 2.10

Consider the element $1 - x \in R[X]$. Is it a unit?

Suppose we have $(1-x)g(x) = 1$, where $g(x) = a_0 + \cdots + a_n x^n$, with $a_n \neq 0$. Then

$$(1-x)g(x) = a_0 + (a_1 - a_0)x + \cdots + (a_n - a_{n-1})x^n - a_n x^{n+1}$$

and then, by comparing the x^{n+1} coefficient, we can see this is not 1.

On the other hand, if we view $1 - x \in R[[X]]$, this has an inverse:

$$(1-x)(1 + x + x^2 + \cdots) = 1$$

In fact, every power series has an inverse as long as r_0 is invertible.

Definition 2.11 (Rings of Laurent series)**.** Write $R[X, X^{-1}]$ for the ring of *Laurent series* with coefficients in R, that is

$$f = \sum_{i \in \mathbb{Z}} a_i x^i$$

with $a_i \in \mathbb{R}$ and all but finitely many $a_i = 0$.

§2.2 Homomorphisms, ideals and quotients

Definition 2.12 (Ring homomorphisms)**.** Let R and S be rings. A function $f : R \to S$ is a *ring homomorphism* if

- $f(r_1 + r_2) = f(r_1) + f(r_2)$;
- $(\implies)$ $f(0) = 0$;
- $f(r_1 r_2) = f(r_1) f(r_2)$;
- $(\not\implies)$ $f(1) = 1$.

The kernel of f is $\ker f = \{r \in R : f(r) = 0\}$, and the image of f is $\operatorname{im} f = \{s \in S : s = f(r) \text{ for some } r \in R\}$.

Definition 2.13 (Ring isomorphisms)**.** If $f : R \to S$ is a bijective ring homomorphism, then it is called an isomorphism. Furthermore, its inverse is also a ring homomorphism.

Definition 2.14. A subset $I \subseteq R$ is called an *ideal*, written $I \lhd R$, if

- I is a subgroup additively;
- if $a \in I$ and $b \in R$, then $ab \in I$.

Usually ideals are not subrings, since they can only contain 1 if $I = R$.

Lemma 2.15

If $\varphi : R \to S$ is a homomorphism, then $\ker \varphi \lhd R$.

Proof. Since φ is a group homomorphism, we know $\ker \varphi$ is a subgroup.
Furthermore, if $a \in \ker \varphi$ and $b \in R$, then $\varphi(ab) = \varphi(a)\varphi(b) = 0 \cdot \varphi(b) = 0$. ∎

Remark. If $a \in R$ is a unit, and $I \lhd R$ is an ideal, then $u \in I \implies I = R$ – since $1 = uv$, $1 \in I$, and so $I = R$.

Example 2.16

Observe that $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$ for any n. In fact, every ideal of $\mathbb{Z}$ has this form. Indeed, if $I \neq \{0\}$, let $n \in \mathbb{Z}$ be the minimal positive element of I. By the Euclidean algorithm, we can write $m \in I$ as $m = qn + r$, with $0 \leq r < n$, and $qn \in I$, so $r \in I$ – this contradicts minimality of n.

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Definition 2.17. Let $A \subseteq R$ be a subset. The *ideal generated by* A is

$$\langle A \rangle = \left\{ \sum_{a \in A} r_a a : r_a \in R, r_a = 0 \text{ for all but finitely many } a \right\}$$

Definition 2.18. An ideal is called *principal* if there exists $r \in R$ such that $I = \langle r \rangle$.

Example 2.19

Consider the subset of $R[x]$ given by

$$\{f(x) \in R(x) : f(0) = 0\}$$

which is generated by $x \in R[x]$ – hence, this is a principal ideal.

Definition 2.20 (Quotients)**.** Let $I \lhd R$ be an ideal. The *quotient ring* R/I is the set of cosets of I in R with 0 and 1 given by $0_R + I$ and $1_R + I$, with the rules

$$(r_1 + I) + (r_2 + I) = (r_1 + r_2) + I$$
$$(r_1 + I) \cdot (r_2 + I) = r_1 r_2 + I$$

Proposition 2.21

The quotient ring is a ring. Furthermore, the function

$$\begin{aligned} f : R &\to R/I \\ r &\mapsto r + I \end{aligned}$$

is a ring homomorphism.

Proof. This is obviously a group homomorphism. Furthermore, multiplication is well-defined: suppose $r_1 + I = r_1' + I$, $r_2 + I = r_2' + I$. We then have $r_i - r_i' \in I$, say $r_i - r_i' = a_i$. We then have

$$r_1' r_2' = (r_1 + a_1)(r_2 + a_2) = r_1 r_2 + \underbrace{r_1 a_2 + r_2 a_1 + a_1 a_2}_{\in I}$$

We conclude that the coset in I of $r_1' r_2'$ is the same as that of $r_1 r_2$.

To check the other ring properties is straightforward. ∎

Example 2.22

- We have $n\mathbb{Z} \lhd \mathbb{Z}$. The quotient ring $\mathbb{Z}/n\mathbb{Z}$ consists of cosets of $n\mathbb{Z}$, i.e. $0 + n\mathbb{Z}, 1 + n\mathbb{Z}, \ldots, (n-1) + n\mathbb{Z}$. The ring structure is modular arithmetic.
- Take $\langle x \rangle \lhd \mathbb{C}[x]$. The elements of the quotient $\mathbb{C}[x]/\langle x \rangle$ can be represented by elements of the form $(a_0 + a_1 x + \cdots + a_n x^n) + \langle x \rangle$ – but all the terms $a_1 x, \ldots, a_n x^n$ lie in $\langle x \rangle$, so every element is uniquely represented by a_0. Hence $\mathbb{C}[X]/\langle x \rangle \cong \mathbb{C}$.

Proposition 2.23 (Euclidean algorithm for polynomial rings over fields)

Let K be a field, and take $f, g \in K[x]$. There exist polynomials $r, q \in K[x]$ such that

$$f = gq + r$$

and $\deg r < \deg g$.

Proof. Let $n = \deg f$ and $m = \deg g$, so

$$f = \sum_{i=0}^{n} a_i x^i \qquad g = \sum_{i=0}^{m} b_i x^i$$

with $a_i, b_i \in K$, $a_n \neq 0$. If $n < m$, set $q = 0$ and $r = f$. On the other hand, if $n \geq m$, we proceed by induction on n. In particular, let $f_1 = f - a_n b_m^{-1} x^{n-m}$ (note we are using the fact that K is a field). Now, $\deg f_1 < n$. If $n = m$, we can write

$$f = (a_n b_m^{-1} x^{n-m}) g + f_1$$

Otherwise, if $n > m$ we can write by induction

$$f_1 = g q_1 + r_1$$

with $\deg r_1 < m$. Then

$$f = (a_n bm^{-1} x^{n-m} + q)g + r_1$$

$\blacksquare$

Corollary

All ideals in $K[x]$ for K a field are principal.

Proof. We have proven that Euclidean algorithm $\implies$ all ideals are principal, since we did it for $\mathbb{Z}$. The result follows. $\blacksquare$

Theorem 2.24 (First isomorphism theorem for rings)

Let $\varphi : R \to S$ be a ring homomorphism. Then, the map

$$\begin{aligned} f : R/\ker\varphi &\to \operatorname{im}\varphi \\ r + \ker\varphi &\mapsto \varphi(r) \end{aligned}$$

is a well-defined ring isomorphism.

Proof. Well-definedness, bijectivity and additivity all follow from the group isomorphism theorem. It remains to check multiplicativity. Observe

$$\begin{aligned} f((r + \ker\varphi)(t + \ker\varphi)) &= f(rt + \ker\varphi) \\ &= \varphi(rt) \\ &= \varphi(r)\varphi(t) \\ &= f(r + \ker\varphi)f(t + \ker\varphi) \end{aligned}$$

$\blacksquare$

Example 2.25

Consider the homomorphism

$$\begin{aligned} \varphi : \mathbb{R}[x] &\to \mathbb{C} \\ f(x) &\mapsto f(i) \end{aligned}$$

The map is clearly surjective, and the kernel is exactly those polynomials f satisfying $f(i) = 0$. However, since coefficients of f are real, $-i$ is also a root. In conclusion, $x^2 + 1 \mid f$, and we conclude $\ker\varphi = \langle x^2 + 1 \rangle$. Hence, by the first isomorphism theorem, $\mathbb{C} \cong \mathbb{R}[x]/\langle x^2 + 1 \rangle$.

Proposition 2.26 (Second isomorphism theorem for rings)
Let $R \leq S$ and $J \lhd S$. Then $J \cap R$ is an ideal in R, and

$$(R+J)/J = \{r+J : r \in R\} \leq S/J$$

is a subring. Furthermore,

$$R/(R \cap J) \cong (R+J)/J$$

Proof. Define a function

$$\begin{aligned} \varphi : R &\to S/J \\ r &\mapsto r+J \end{aligned}$$

so $\ker \varphi = \{r \in R : r+J = 0\} = R \cap J$, so this is an ideal by the first isomorphism theorem. Also, the image is $\operatorname{im} \varphi = \{r+J : r \in R\} = (R+J)/J$. The result follows from the first isomorphism theorem. ∎

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Theorem 2.27 (Correspondence between subrings and ideals)
If $I \lhd R$, there exists a bijection

$$\begin{aligned} \{\text{subrings of } R \text{ containing } I\} &\leftrightarrow \{\text{subrings of } R/I\} \\ I \lhd S \leq R &\mapsto S/I \leq R/I \\ L \leq R/I &\mapsfrom \{r \in R \mid r+I \in L\} \end{aligned}$$

Similarly, there is a bijection between ideals of R/I and ideals of R containing I.

Remark. This is powerful, because it allows us to talk about the structure of quotient rings using only information about the original ring.

Theorem 2.28 (Third isomorphism theorem for rings)
Let I, $J \lhd R$ with $I \subseteq J$. Then $J/I \lhd R/I$, and $(R/I)/(J/I) \cong R/J$.

Proof. Define a function

$$\begin{aligned} \phi : R/I &\to R/J \\ r+I &\mapsto r+J \end{aligned}$$

which is well-defined by the groups result. It is easy to check that this is a ring homomorphism. Have

$$\ker \phi = \{r+I \mid r+J = 0\} = J/I$$

so the result follows by the first isomorphism theorem. ■

Proposition 2.29

Let R be a ring. There is a *unique* homomorphism $i : \mathbb{Z} \to R$ sending 1 to 1_R. Then $\ker i \triangleleft \mathbb{Z}$, so $\ker i = n\mathbb{Z}$ for some $n \in \mathbb{Z}$ – n is called the *characteristic* of R.

Remark. Why does this work? The integers are a special ring for which addition and multiplication are particularly closely related. Sometimes, $\mathbb{Z}$ is called the *initial* ring.

Recall that, in group theory, there exists a homomorphism from the trivial group to any other group. In ring theory, this is not the case – the existence of a homomorphism from the zero ring would imply $1 = 0$. Instead, $\mathbb{Z}$ plays this role.

Example

$\mathbb{Z}$, $\mathbb{R}$, $\mathbb{C}$ have characteristic zero. $\mathbb{Z}/k\mathbb{Z}$ has characteristic k.

§2.3 Integral domains, fields of fractions, and prime ideals

Notice that, in $\mathbb{Z}/\langle 6\rangle$, $2 \cdot 3 = 0$, while $2, 3 \neq 0$.

Definition 2.30 (Zero divisor)**.** A *zero divisor* in a ring R is a non-zero a such that there exists $b \neq 0$ with $a \cdot b = 0$.

Definition 2.31 (Integral domain)**.** A non-zero ring R is an *integral domain* if R lacks zero divisors – for all $a, b \in \mathbb{R}$, $a \cdot b = 0 \implies a = 0$ or $b = 0$.

Example

$\mathbb{Z}$, $\mathbb{Z}[i]$, $\mathbb{Q}$, $\mathbb{C}$, $\mathbb{R}[x]$, $\mathbb{Z}[x]$ are all integral domains. On the other hand, $\mathbb{Z}/\langle 6\rangle$, $\mathbb{Z}/\langle pq\rangle$, $\mathbb{Z}/\langle 2\rangle \times \mathbb{Z}/\langle 2\rangle$ are all not integral domains.

Any field is an integral domain, because, if $ab = 0$ with $b \neq 0$, then $a = abb^{-1} = 0$. In fact, we can say something stronger – all subrings of fields are integral domains. On the other hand, products of rings are clearly never integral domains.

Lemma 2.32

Let R be a *finite* integral domain. Then R is a field.

Proof. Let $a \in R$ be non-zero. Define

$$\begin{aligned} \mu_a : R &\to R \\ r &\mapsto ar \end{aligned}$$

which is a group homomorphism. Since R is an integral domain, $\ker \mu_a = \{0\}$. Hence, μ_a is injective. Since R is finite, μ_a is also surjective, so $1 \in \operatorname{im} \mu_a$. The result follows. ■

Definition 2.33. Let R be an integral domain. A *field of fractions* of R is a field F satisfying the following two properties:

- $R \leq F$ is a subring;
- every $x \in F$ can be written as ab^{-1}, for a, $b \in R$ and b^{-1} the multiplicative inverse of b when viewed as an element of F.

Example
$\mathbb{Q}$ is a field of fractions of $\mathbb{Z}$.

Theorem 2.34
Every integral domain R has a field of fractions.

Proof. Define

$$S = \{(a, b) \in R \times R \mid b \neq 0\}$$

and an equivalence relation $\sim$ such that

$$(a, b) \sim (c, d) \iff ad = bc \text{ in } R$$

That $\sim$ is symmetric and reflexive are clear, but transitivity is not quite so trivial. Suppose $(a, b) \sim (c, d)$, $(c, d) \sim (e, f)$, so $ad = bc$, $cf = de$. Multiplying the first equality by f and the second by b, we obtain

$$\begin{aligned} adf &= bcf = bde \\ \implies d(af - be) &= 0 \end{aligned}$$

Since $d \neq 0$ by hypothesis, and R is an integral domain, obtain $af = be$, so $(a, b) \sim (e, f)$ as required.

Now, define $F = S/\sim$. As notation, write $\frac{a}{b} = [(a, b)]$. We make F into a ring with the following well-defined operations:

$$\begin{aligned} \frac{a}{b} + \frac{c}{d} &= \frac{ad + bc}{bd} \\ \frac{a}{b} \cdot \frac{c}{d} &= \frac{ac}{bd} \end{aligned}$$

To see that F is a field, let $\frac{a}{b} \neq 0_F$, so $\frac{a}{b} \neq \frac{0}{1}$, so $a \cdot 1 \neq b \cdot 0$, so $a \neq 0$. Hence, we have $\frac{a}{b}^{-1} = \frac{b}{a}$ as required.

Finally, define

$$\begin{aligned} \phi : R &\to F \\ r &\mapsto \frac{r}{1} \end{aligned}$$

which is clearly a ring homomorphism. Have $\ker \phi = \{r \in R \mid \frac{r}{1} = 0\} = \{0\}$. By the first isomorphism theorem, have $\operatorname{im} \phi \cong R$, and

$$\frac{a}{b} = \frac{a}{1} \cdot \frac{1}{b} = \frac{a}{1} \cdot \frac{b}{1}^{-1}$$

as required. ∎

> **Remark.** It follows that every integral domain is a subring of some field.

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Proposition 2.35

Let R be a ring. Then R is a field if and only if the only ideals of R are $\{0\}$ and R.

Proof. $\implies$ If R is a field and $I \lhd R$, say $I \neq \{0\}$, then I contains a unit, say $u \in I$. But then $1 \in I$ since $1 = uv$. We conclude $I = R$.

$\impliedby$ If the only ideals are $\{0\}$ and R, take $0 \neq r \in R$. Then $\langle r \rangle$ is R. So $1 \in R \implies 1 = rs$. So r is a unit and hence R is a field.

■

Definition 2.36 (Maximal ideals)**.** An ideal $R \neq I \lhd R$ is called *maximal* if, for any $J \lhd R$, $I \subseteq J \subseteq R$ implies $I = J$ or $J = R$.

Proposition 2.37

An ideal $I \lhd R$ is maximal if and only if R/I is a field.

Proof. R/I is a field if and only if its only ideals are $\{0\}$ and R/I itself. Now apply the ideal correspondence result. ■

Definition 2.38 (Prime ideals)**.** An ideal $R \neq I \lhd R$ is *prime* if, whenever $ab \in I$, either a or $b \in I$.

Example 2.39

An ideal $n\mathbb{Z} \lhd \mathbb{Z}$ is a prime ideal if and only if n is 0 or a prime.

Proposition 2.40

An ideal $I \lhd R$ is prime if and only if R/I is an integral domain.

Proof. $\implies$ Suppose I is prime and let $a + I$ and $b + I \in R + I$ be cosets. Suppose $(a + I)(b + I) = ab + I = 0 + I$ if and only if $ab \in I$. But, since I is prime, either a or $b \in I$, so either $a + I$ or $b + I = 0 + I$.

$\impliedby$ If R/I is an integral domain, then $ab \in I$ satisfies $ab + I = (a + I)(b + I) = 0 + I$ in R/I, so either $a \in I$ or $b \in I$, so I is prime.

■

Corollary 2.41

Every maximal ideal in a ring is prime.

Proof. Since every field is an integral domain, the result follows from proposition 2.37 and proposition 2.40. ∎

Remark. The converse clearly does not hold.

Proposition 2.42

The characteristic of an integral domain is either zero or a prime number.

Proof. Consider the unique homomorphism $\mathbb{Z} \to R$. Letting the characteristic be n, we have that $\langle n \rangle \triangleleft \mathbb{Z}$ is the kernel, so $\mathbb{Z}/n\mathbb{Z} \leq R$ by the first isomorphism theorem. But then $\mathbb{Z}/n\mathbb{Z}$ is an integral domain, and the result follows. ∎

§2.4 Factorisation in integral domains

Throughout this section, R is an integral domain.

Definition 2.43 (Division)**.** Let $a, b \in R$. We say a *divides* b, written $a \mid b$, if there exists $c \in R$ such that $ac = b$. Equivalently, we can say $\langle b \rangle \subseteq \langle a \rangle$.

Definition 2.44 (Associates)**.** We say $a, b \in R$ are *associates* if $a = bc$ for some unit $c \in R$. Equivalently, we can say $a \mid b$ and $b \mid a$, or $\langle a \rangle = \langle b \rangle$.

Definition 2.45 (Irreducible elements)**.** An element $r \in R$ is *irreducible* if $r \neq 0$, r is not a unit, and, if $r = xy$, either x or y is a unit.

Definition 2.46 (Prime elements)**.** An element $r \in R$ is *prime* if $r \neq 0$, r is not a unit, and, if $r \mid xy$, either $r \mid x$ or $r \mid y$.

Proposition 2.47

Let $r \in R$. Then, if $r \neq 0$, it is a prime element if and only if $\langle r \rangle$ is a prime ideal.

Proof. $\implies$ Let r be a prime element, and suppose $ab \in \langle r \rangle$. Then $r \mid ab$, so $r \mid a$ or $r \mid b$, hence $a \in \langle r \rangle$ or $b \in \langle r \rangle$ as required.

$\impliedby$ Let $\langle r \rangle$ be a prime ideal. By definition, r is not a unit. Say $r \mid ab$, so $ab \in \langle r \rangle$. Since $\langle r \rangle$ is prime, either $a \in \langle r \rangle$ or $b \in genr$, so either $r \mid a$ or $r \mid b$.

∎

Proposition 2.48

Let $r \in R$ be prime. Then r is irreducible.

Proof. Let $r \in R$ be prime, and suppose $r = ab$. Since $r \mid r$, $r \mid ab$, so, since r is prime, $r \mid a$ or $r \mid b$ – wlog suppose $r \mid a$. Hence $a = rc$ with $c \in R$, so $r = ab = rcb$. Since we're in an integral domain, we can cancel to get $bc = 1$, so b is a unit as required. ■

> **Remark.** The converse is false – for example, $R = \mathbb{Z}[\sqrt{-5}]$.

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Example

Take $R = \mathbb{Z}[\sqrt{-5}] \leq \mathbb{C}$, which is clearly an integral domain. There exists on this ring a *norm*

$$\begin{aligned} N\colon \mathbb{Z}[\sqrt{-5}] &\to \mathbb{Z}_{\geq 0} \\ a + b\sqrt{-5} &\mapsto a^2 + 5b^2 \end{aligned}$$

Observe that this function is multiplicative. Such a norm is useful for classifying units – supposing $ab = 1$, we have $N(a)N(b) = N(1) = 1$, so $N(a) = N(b) = 1$. Hence, the only units in R are ± 1.

Furthermore, we can show 2 is irreducible – have $N(2) = 4$, and there is clearly no element of norm 2, so $2 = ab \implies a$ or b a unit. Similarly, 3, $1 \pm \sqrt{-5}$ are all irreducible. Yet –

$$2 \cdot 3 = 6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

Observing that $4 = N(2) \nmid N(1 \pm \sqrt{-5}) = 6$, we conclude that 2 is not prime. Hence, we have an irreducible which is not prime. Now, unique factorisation also fails.

Definition 2.49 (Euclidean domain)**.** An integral domain R is an *Euclidean domain* if there exists a function

$$\phi : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$$

such that

- $\phi(ab) \geq \phi(b)$ for all non-zero a, b;
- if a, $b \in R$ with $b \neq 0$, there exist q, $r \in R$ with $a = bq + r$ such that either $r = 0$ or $\phi(r) < \phi(b)$.

Example

Examples include $\mathbb{Z}$ with $\phi = |\cdot|$ and $K[x]$ with K a field, $\phi = \deg$. $\mathbb{Z}[\sqrt{-5}]$, on the other hand, is *not* a Euclidean domain.

Example

The Gaussian integers $R = \mathbb{Z}[i] \leq \mathbb{C}$ are a Euclidean domain. Indeed, define

$$\begin{aligned} \phi : \mathbb{Z}[i] &\to \mathbb{Z}_{\geq 0} \\ z &\mapsto |z|^2 \end{aligned}$$

We claim that ϕ makes R Euclidean, which requires us to check the two properties. The first one is obvious. Let us attack the second. Take a, $b \in \mathbb{Z}[i]$, $b \neq 0$. Consider the ratio $a/b \in \mathbb{C}$. There is a point $q \in \mathbb{Z}[i]$ such that

$$\left|\frac{a}{b} - q\right| < 1$$

Hence, $\frac{a}{b} = q + c$, where $|c| < 1$, giving $a = bq + bc$. Taking $r = bc$, we see

$$\phi(r) = \phi(bc) = \phi(b)\phi(c) = \phi(b)|c|^2 < \phi(b)$$

as required.

Remark. This is pretty easy to generalise – all we needed was that $\left|\frac{a}{b} - q\right| < 1$ for a suitable choice of q.

Remark. By the same proof as for $\mathbb{Z}$, we conclude that every ideal in $\mathbb{Z}[i]$ is principal.

Definition 2.50 (Principal ideal domain)**.** A *principal ideal domain (PID)* is an integral domain R such that every ideal $I \lhd R$ is principal.

Proposition 2.51

Every Euclidean domain is a PID.

Proof. Replicate the proof for $\mathbb{Z}$, taking the element of the ideal with smallest size w.r.t. the Euclidean function. ∎

Example

Let $R = \mathbb{Z}[x]$ – we claim this is neither a Euclidean domain nor a principal ideal domain. Take $I = \langle 2, x\rangle$, which we claim is not principal. Indeed, if $I = \langle f\rangle$, have $f \mid 2$, so $f \in \{\pm 1, \pm 2\}$. If $f = \pm 1$, have $\langle f\rangle = \mathbb{Z}[x] \neq \langle 2, x\rangle$ clearly. Hence, $f = \pm 2$, but $2 \nmid x$.

Definition 2.52 (Unique factorisation domain)**.** A *unique factorisation domain (UFD)* is an integral domain R such that

- (existence) every non-unit in R is a product of finitely many irreducibles;
- (uniqueness) if $p_1 \dots p_n = q_1 \dots q_m$ with p_i, q_j all irreducible, then $n = m$ and, up to reordering, p_i and q_i are associates for all i.

Lemma 2.53

In a PID, every irreducible is prime.

Proof. Let $p \in R$ be irreducible. Suppose $p \mid ab$, $p \nmid a$. Consider $\langle p, a \rangle \lhd R$, which is principal, so is equal to $\langle d \rangle$ for some d. Then $d \mid a$, $d \mid p$. But $d \mid p \implies p = q_1 d$ which, by irreducibility, implies either q_1 or d a unit. If q_1 is a unit, then d and p are associates, so $d \mid a \implies p \mid a$, contradiction.

Hence, $\langle d \rangle = \langle 1 \rangle = R$. Therefore, $1 = rp + sa$ for some r, $s \in R$. But then

$$b = rpb + sab$$

where $p \mid \text{RHS}$. The result follows. ∎

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Theorem 2.54 (PID $\implies$ UFD)

Any PID is a UFD.

Remark. The converse is false – take $\mathbb{C}[x, y]$.

Lemma 2.55

Let R be a PID. If $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$ is a chain of ideals in R, there exists $N > 0$ such that, for all $n \geq N$, $I_n = I_N$.

Remark. In $\mathbb{Z}$, this follows from the fact that every decreasing sequence of naturals is eventually constant.

Proof. Take

$$I = \bigcup_j I_j$$

which is a principal ideal by assumption, so $I = \langle r \rangle$. But then there exists some N with $r \in I_N$, after which point the sequence clearly stabilises. ∎

Proof of theorem. First, we show existence. Let $r \in R$. If r is irreducible, we are done. Assuming not, write $r = r_1 s_1$, where r_1, s_1 are not units. If r_1, s_1 are products of irreducibles, we are done. Otherwise, without loss of generality, say r_1 is not a product of irreducibles. Continuing, we obtain a sequence $r, r_1, \dots$, which gives

$$\langle r \rangle \subseteq \langle r_1 \rangle \subseteq \dots$$

Now, by the lemma, this stabilises, so have some N such that, for $n \geq N$, $\langle r_n \rangle = \langle r_{n+1} \rangle = \langle r_{n+1} s_{n+1} \rangle$. Then s_{n+1} is a unit, contradiction.

It remains to show uniqueness. Suppose $p_1 \dots p_n = q_1 \dots q_m$. Now $p_1 \mid q_1 \dots q_m$, so, since p_1 is prime (PID), $p_1 \mid q_i$ for some i. Then, since q_i is irreducible, have $q_i = ap_1$, a a unit. Relabelling for simplicity, take $i = 1$, so we have $p_1 \cdots p_n = ap_1 \cdots q_m$ so, cancelling, have $p_2 \cdots p_n = (aq_2) \cdots q_m$. Replacing q_2 with aq_2, we can continue this procedure, to obtain that p_i, q_i are associates for every i (and $n = m$). ■

> **Remark.** In any UFD, irreducible $\implies$ prime.

Definition 2.56 (GCD and LCM)**.** Let $a_1, \dots, a_n \in R$. We say that d is an *greatest common divisor (GCD)* if

- $d \mid a_i$ for each i;
- if $d' \mid a_i$ for all i, $d' \mid d$.

Similarly, we say that m is an *least common multiple (LCM)* if

- $a_i \mid m$ for each i;
- if $a_i \mid m'$ for each i, $m \mid m'$.

Theorem 2.57 (Existence of GCD and LCM)

Let R be a UFD, then GCDs and LCMs exist and are unique up to associates.

Proof. Let $p_1, \dots, p_m$ be all irreducible factors in the unique factorisations of all the a_i (up to associates). Hence, we can write

$$a_i = u_i \prod_{j=1}^{m} p_j^{n_{ij}}$$

where $0 \leq n_{ij}$ and u_i is a unit. Let $m_j = \min_i n_{ij}$, and put

$$d = \prod_{j=1}^{m} p_j^{m_j}$$

By definition, $m_j \leq n_{ij}$, so $d \mid a_i$ for each i. For the greatest part, suppose $d' \mid a_i$ for all i, then

$$d' = v \prod_{j=1}^{m} p_j^{t_j}$$

for v a unit, so we must have $t_j \leq m_j$ for all j. Hence, we have existence. Uniqueness is immediate – if $d' \mid d$ and $d \mid d'$, d' and d are associates. LCM is analogous. ■

§2.5 Factorisation in polynomial rings

Recall that, if k is a field, then $k[x]$ has the following properties:

- $k[x]$ is Euclidean;
- every $I \lhd k[x]$ is principal;
- $f \in k[x]$ is irreducible $\iff$ f prime.

Now, let f be irreducible, and take $\langle f \rangle \lhd k[x]$, which is clearly a prime ideal. Suppose $\langle f \rangle \subseteq J \lhd k[x]$, then $J = \langle g \rangle$, so $f = gh$, and hence either g and f are associates or g is a unit. Hence, $\langle f \rangle$ is maximal. In particular, f irreducible if and only if $k[x]/\langle f \rangle$ is a field. Now, we work more generally.

Definition 2.58 (Content and primitive polynomials)**.** Let R be a UFD, and $f = a_0 + a_1 x + \cdots + a_n x^n \in R[x]$. The *content* of f

$$c(f) = \gcd(a_0, \ldots, a_n) \in R$$

If $c(f)$ is a unit, we say f is primitive.

Theorem 2.59 (Gauss's lemma)

Let R be a UFD, and $f \in R[x]$. If f is primitive, then f is reducible in $R[x]$ if and only if f is reducible in $F[x]$, where F is the field of fractions of R.

Example

Consider $2x + 2 \in \mathbb{Z}[x]$, which is not primitive. Indeed, it is reducible in $\mathbb{Z}[x]$ but not in $\mathbb{Q}[x]$. This is the only way in which the theorem can fail.

Lecture

Example

Consider the polynomial $f = x^3 + x + 1 \in \mathbb{Z}[x]$. Clearly $c(f) = 1$. We will show that f is irreducible in $\mathbb{Z}[x]$, and conclude it is also irreducible in $\mathbb{Q}[x]$. Supposing for contradiction it were irreducible, have $f = gh$, g, $h \in \mathbb{Z}[x]$ not units. In order for the content to be 1, $\deg g$, $\deg h \geq 1$. Hence, we may take wlog $g = b_0 + b_1 x$, $h = c_0 + c_1 x + c_2 x^2$, giving

$$b_0 c_0 = 1$$
$$b_1 c_2 = 1$$

so b_0, $b_1 = \pm 1$. Hence, ± 1 is a root of g and hence of f. But it is not.

Lemma 2.60

Let R be a UFD. If f, $g \in R[x]$ are primitive, then fg is primitive.

Proof. Let $f = a_0 + \cdots + a_n x^n$, $g = b_0 + \cdots + b_m x^m$, with a_n, $b_m \neq 0$, and suppose $c(f) = c(g) = 1$. Suppose $c(fg)$ is not a unit, so we have an irreducible $p \in R$ with $p \mid c(fg)$. But $p \nmid c(f), c(g)$. Suppose k, l are the smallest indices such that $p \nmid a_k$, $p \nmid b_l$. Then, the coefficient of x^{k+l} in fg is

$$\sum_{i+j=k+l} a_i b_j$$

We have $p \mid a_{k+l}b_0, \ldots, a_{k+1}b_{l-1}, a_{k-1}b_{l+1}, \ldots, a_0 b_{l+k}$ by minimality – but, by assumption, the sum is divisible by p. Hence, $p \mid a_k b_l$ also. Now, by primality of p (R a UFD), have $p \mid a_k$ or b_l, contradiction. ∎

Corollary 2.61

Let R be a UFD, f, $g \in R[x]$. Then $c(fg)$ and $c(f)c(g)$ are associates.

Proof. Write $f = c(f)f_1$, $g = c(g)g_1$ with f_1, g_1 primitive. The result immediately follows by the lemma. ∎

Proof of Gauss's lemma. Take $f \in R[x]$ primitive (R a UFD). If $f = gh$ in $R[x]$ with g, h not units, since f is primitive, g, h primitive, so $\deg g, \deg h > 0$. Hence f is reducible in $F[x]$.

The other direction is not quite so simple. Let $f = gh$ in $F[x]$ with g, h non-units, so $\deg g, \deg h > 0$. Let a, $b \in R[x]$ such that $abf = (ag)(bh)$, where ag, $bh \in R[x]$. (Note that they are not *products* in $R[x]$, since g, h not necessarily elements of $R[x]$). Now write $ag = c(ag)g_1$, $bh = c(bh)h_1$, with g, h primitive. Since f primitive, $c(abf) = ab$ so, by the corollary,

$$ab = c(abf) = c((ag)(bh)) = c(ag)c(bh)u$$

for u a unit. But now

$$abf = c(ag)c(bh)g_1h_1 = u^{-1}abg_1h_1$$

so, cancelling ab, we are done. ∎

Proposition 2.62

Let R a UFD, F its field of fractions. Take $g \in R[x]$ primitive. Then, if $f \in R[x]$ is divisible by g in $F[x]$, it is also divisible by g in $R[x]$.

In other words, if $I = \langle g \rangle \triangleleft R[x]$, $J = \langle g \rangle \triangleleft F[x]$, then $I = J \cap R[x]$.

Proof. Let $f \in J \cap R[x]$, so $f = gh$ in $F[x]$. Multiplying by $b \in R$, have $bf = g(bh)$ in $R[x]$. Setting $bh = c(bh)h_1$ with h_1 primitive, have gh_1 primitive, so $c(bh) = uc(bf)$ for u a unit. But $c(bf) = bc(f)$, since b, $f \in R[x]$. Now observe

$$bf = gubc(f)h_1$$

which gives the result after cancelling b. ∎

Theorem 2.63

If R is a UFD, so is $R[x]$.

Proof. Let $f \in R[x]$. Write $f = c(f)f_1$ with f_1 primitive. Factorise $c(f) \in R$ as $p_1 \dots p_n$, with p_i irreducible. It remains, then, to produce a factorisation of f_1, which we will do by induction on the degree. If f_1 is irreducible, we are done – otherwise, write $f_1 = f_{11}f_{12}$ where f_{11} and f_{12} are necessarily primitive and have smaller degrees. Since the degree is strictly decreasing, the process terminates. Hence, $f_1 = q_1 \dots q_m$, a factorisation into irreducibles, so $f = p_1 \dots p_n q_1 \dots q_m$.

It remains to show uniqueness. The uniqueness of the p_i follows immediately from R being a UFD. Uniqueness for the q_i is as follows: consider two factorisations $f_1 = q_1 \dots q_m = r_1 \dots r_l$. We can view this not just in $R[x]$ but in $F[x]$, which is a Euclidean domain hence a UFD. Thus, $l = m$ and, up to reordering, the r_i and q_i are associates in $F[x]$. Then proposition 2.62 gives that r_i and q_i are associates in $R[x]$ also, as required. ■

Remark. This gives many examples of UFDs which are not PIDs.

Lecture

Theorem 2.64 (Eisenstein's criterion)

Take R a UFD, $f = a_0 + \dots + a_n x^n \in R[x]$ primitive with $a_n \neq 0$. Suppose there exists $p \in R$ irreducible such that

- $p \nmid a_n$;
- $p \mid a_i$ for $0 \leq i < n$;
- $p^2 \nmid a_0$.

Then f is irreducible in $R[x]$ or indeed $F[x]$.

Proof. Suppose $f = gh$ with $g = r_0 + \dots + r_k x^k$, $h = s_0 + \dots + s_l x^l$, r_k, $s_l \neq 0$. Since $p \nmid a_n$, know $p \nmid r_k$, s_l. Also, p divides *either* r_0 or s_0, but not both – without loss of generality, suppose $p \mid r_0$. Let j be the index such that

$$p \mid r_0,\, p \mid r_1, \dots, p \mid r_{j-1},\, p \nmid r_j$$

Now,

$$a_j = r_0 s_j + r_1 s_{j-1} + \dots + r_j s_0$$

Then, $p \nmid s_0$, $p \nmid r_j \implies p \nmid a_j$. Hence, we must have $j = n$, and so furthermore $k = n$. Hence $\deg g = n$, $\deg h = 0$ so, since f primitive, $f = gh$ is not a proper factorisation. ■

Example

Let us work with $R = \mathbb{Z}$, and $f = x^n - p$, p a prime. The irreducibility of f is immediate – since this extends to $\mathbb{Q}$, we conclude $\sqrt[n]{p}$ is irrational.

Example

Fix p a prime, take $f(x) = x^{p-1} + x^{p-2} + \cdots + x + 1 = \frac{x^p - 1}{x-1}$. Eisenstein's criterion does not immediately apply – but it does apply to

$$\hat{f}(y) = f(y+1) = y^{p-1} + \binom{p}{1} y^{p-2} + \cdots + \binom{p}{p-1}$$

Now Eisenstein applies to show $\hat{f}$ is irreducible, whence the irreducibility of f follows.

§2.6 Gaussian integers

The ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\} \leq \mathbb{C}$. It has a norm

$$\begin{aligned} N \colon \mathbb{Z}[i] &\to \mathbb{Z}_{\geq 0} \\ a + bi &\mapsto a^2 + b^2 \end{aligned}$$

This norm enables division with smaller remainder, making $\mathbb{Z}[i]$ into a Euclidean domain, hence PID, hence UFD.

Remark. Let us make some observations.

- $\alpha \in \mathbb{Z}[i]$ is irreducible iff it is prime;
- the units in $\mathbb{Z}[i]$ have norm 1, namely ± 1, $\pm i$.
- $2 = (1+i)(1-i)$, so 2 is not prime in $\mathbb{Z}[i]$.
- 3 is prime, since there are no elements of norm 3.

Proposition 2.65

A prime $p \in \mathbb{Z}$ remains prime in $\mathbb{Z}[i]$ if and only if $p \neq a^2 + b^2$ for $a, b \in \mathbb{Z} \setminus \{0\}$.

Proof. If $p = a^2 + b^2$, $p = (a + bi)(a - bi)$, so p is not irreducible. Conversely, suppose $p = uv$ with u, v not units in $\mathbb{Z}[i]$. Then

$$p^2 = N(p) = N(u)N(v)$$

so $N(u) = N(v) = p$. But then, taking $u = a + bi$ gives the result. ∎

Theorem 2.66 (Primes in the Gaussian integers)

The primes in $\mathbb{Z}[i]$ (up to associates) fall into the following classes:

- the primes $p \equiv 3 \pmod 4$ in $\mathbb{Z}$;
- $z \in \mathbb{Z}[i]$ with $N(z) = p$ such that $p \equiv 1 \pmod 4$, or $p = 2$.

Lemma 2.67

Let p be a prime, $\mathbb{F}_p$ the field with p elements. Then the group $\mathbb{F}_p^\times$ is cyclic.

Proof. Clearly $\mathbb{F}_p^\times$ is abelian with order $p-1$. By the classification of finite abelian groups, if $\mathbb{F}_p^\times$ is not cyclic, it contains as a subgroup $C_m \times C_m$ with $m > 1$. Hence, there are at least m^2 elements in $\mathbb{F}_p$ such that $x^m - 1 = 0$. But such a polynomial can have at most m roots over a field – contradiction. ■

Proof of theorem. There are many steps

$\Longleftarrow$ If $p \equiv 3 \pmod 4$, then $p \neq a^2 + b^2$, because a sum of two squares is never 3 mod 4. If $N(z) = p$, $z = uv$, then $N(u)N(v) = p$. Hence, either u or v has norm 1, so z is irreducible.

$\Longrightarrow$ Let $z \in \mathbb{Z}[i]$ a prime. Then, $\bar{z}$ is also irreducible, so $N(z) = z\bar{z}$ is a factorisation of the norm into irreducibles. Suppose $p \mid N(z)$, p a prime in $\mathbb{Z}$. If $p \equiv 3 \pmod 4$, know p prime in $\mathbb{Z}[i]$, so $p \mid z$ or $p \mid \bar{z}$. Since p is fixed by conjugation, we may assume without loss of generality $p \mid z$. Then, z prime gives $z = p$.

If $p \equiv 1 \pmod 4$, $p = 2$, want to show $N(z) = p$. First, we claim that p is not a prime in $\mathbb{Z}[i]$. If $p \equiv 1 \pmod 4$, have $p - 1 = 4k$, $k \in \mathbb{Z}$. By the lemma, it follows $\mathbb{F}_p^\times \cong C_{4k}$, so there is a unique element in $\mathbb{F}_p^\times$ of order 2, namely -1. We also know that there exists $a \in \mathbb{F}_p^\times$ of order 4. Hence, $a^2 \equiv -1 \pmod 4$, so $p \mid a^2 + 1$. Thus, $p \mid (a+i)(a-i)$ – but clearly $p \nmid a + i$, $p \nmid a - i$, so p is not irreducible.

Thus, can have $p = z_1 z_2$, where neither are units. Observe $p^2 = N(z_1)N(z_2)$, where neither $N(z_1)$, $N(z_2)$ are units, so $N(z_1)$, $N(z_2) = p$. Then

$$p = z_1\bar{z}_1 = z_2\bar{z}_2 = z_1 z_2$$

Thus, $\bar{z}_1 = z_2$ and vice versa. Finally, then,

$$p = z_1\bar{z}_1 \mid N(z) = z\bar{z}$$

By assumption, z_1, $\bar{z}_1$, z, $\bar{z}$ are all irreducible, so either z, z_1 or z, $\bar{z}_1$ are associates. Thus $N(z) = N(z_1) = p$ as required.

■

Remark. The proof gives a way to find $\alpha \in \mathbb{Z}[i]$ such that $N(\alpha) = p$ whenever $p \equiv 1 \pmod 4$.

Corollary 2.68

An integer n can be written as a sum of squares if and only if

$$n = \prod p_i^{n_i}$$

satisfies n_i even whenever $p_i \equiv 3 \pmod 4$.

Proof. $\implies$ If $n = x^2 + y^2$, have $n = N(x+iy)$ for $x + iy \in \mathbb{Z}[i]$. Factorising $z = x + iy$ into Gaussian primes gives $z = \alpha_1 \cdots \alpha_q$. Then, for each i, either α_i is 3 mod 4 or $N(\alpha_i)$ is a prime which is 1 mod 4, or 2. Then

$$x^2 + y^2 = N(z) = \prod N(\alpha_i)$$

and each $N(\alpha_i)$ is p^2 if $p \equiv 3 \pmod 4$.

$\impliedby$ Conversely, let $n = \prod p_i^{n_i}$ satisfying the given conditions. In this case, if $p_i \equiv 3 \pmod 4$,

$$p_i^{n_i} = \left(p_i^{n_i/2}\right)^2 = N(p_i^{n_i/2})$$

If $p \equiv 1 \pmod 4$ or 2, then $p_i = N(\alpha_i)$ from the proof of the theorem. Hence n is the norm of some Gaussian integer and we are done.

■

Example

Take $n = 65 = 13 \cdot 5$. Following the proof, we can write

$$13 = (2+3i)(2-3i) = N(2+3i) \qquad 5 = (2+i)(2-i) = N(2+i)$$

Then

$$65 = N\left((2+3i)(2+i)\right) = N(1+8i)$$

as required. Picking other pairs of factors, we could get other expressions – $7^2 + 4^2$, for example.

§2.7 Algebraic integers

Definition 2.69 (Algebraic integer)**.** A complex number α is an *algebraic integer* if it is the root of a monic polynomial in $\mathbb{Z}[x]$.

We write

$$\mathbb{Z}[\alpha] = \bigcap_{\substack{R \leq \mathbb{C} \\ \alpha \in R}} R$$

or, equivalently, the image of the homomorphism

$$\begin{aligned} e_\alpha : \mathbb{Z}[x] &\to \mathbb{C} \\ g(x) &\mapsto g(\alpha) \end{aligned}$$

Proposition 2.70

Let α be an algebraic integer. The ideal $I = \ker e_\alpha \lhd \mathbb{Z}[x]$ is principal. It is equal to $\langle f_\alpha \rangle$, with f_α irreducible and monic.

Proof. By definition, $I = \ker e_\alpha$ is nonzero. Take $f_\alpha \in I$ of minimal degree. We can assume that f_α is primitive, since α is the root of some monic polynomial. We claim that $I = \langle f_\alpha \rangle$. Let $h \in I$. In $\mathbb{Q}[x]$, we can run the division algorithm, to obtain

$$h = f_\alpha q + r \qquad\qquad \deg r < \deg f_\alpha$$

Clearing denominators, we obtain $ah = af_\alpha q + ar$, $a \in \mathbb{Z}$. Plugging in α, we obtain $0 = 0 + ar(\alpha)$. Hence $r \in I$, so $r = 0$. Therefore, $ah = af_\alpha q$. Taking contents and recalling f_α primitive, we have

$$c(ah) = c(aq)$$

Then, because $h \in \mathbb{Z}[x]$, $a \mid c(ah)$, so $a \mid c(aq)$. Therefore, can write $aq = a\bar{q}$, where $\bar{q} \in \mathbb{Z}[x]$. Thus $q = \bar{q} \in \mathbb{Z}[x]$, so $f_\alpha \mid h$ as required.

To see f_α irreducible, observe $\mathbb{Z}[x]/\langle f_\alpha \rangle = \mathbb{Z}[x]/\ker(e_\alpha) \cong \operatorname{im}(e_\alpha) \leq \mathbb{C}$, so $\mathbb{Z}[x]/\langle f_\alpha \rangle$ is an integral domain. The result follows. ■

Example

Algebraic integers include $\sqrt{2}$, i, $\sqrt{-3}$. What about $\frac{1+\sqrt{-3}}{2}$? Yes, it is a root of unity! So, somehow, we have ended up with an algebraic *integer* which looks like a fraction.

Proposition 2.71

Let $\alpha \in \mathbb{C}$ be an algebraic integer. If $\alpha \in \mathbb{Q}$, then $\alpha \in \mathbb{Z}$.

Proof. Let f_α be the polynomial generating $\ker e_\alpha$. This is irreducible over $\mathbb{Z}[x]$, and thus $\mathbb{Q}[x]$. But $x - \alpha \mid f_\alpha$ in $\mathbb{Q}[x]$, so $f_\alpha = x - \alpha$. The result follows. ■

Lecture

Lecture

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§2.8 Hilbert's basis theorem

Definition 2.72 (Noetherian ring)**.** A ring R is *Noetherian* if every chain $I_1 \subseteq I_2 \subseteq \cdots$ of ideals in R stabilises, i.e. there exists $N \geq 0$ such that $I_n = I_N$ for all $n \geq N$.

> **Remark.** We also have a corresponding *ascending* chain condition, but it is much less important.

Proposition 2.73

A ring R is Noetherian if and only if all ideals are finitely generated.

Proof. Suppose all a ring's ideals are finitely generated, and we are given $I_1 \subseteq I_2 \subseteq \cdots$ a chain of ideals. Consider

$$I = \bigcup I_n$$

This is an ideal, finitely generated by, say, $r_1, \ldots, r_k$. But there is an N such that $r_i \in I_N$ for all i, so, for $j \geq N$, $I_j = I_N$.

Conversely, if I is not finitely generated, pick $r_1 \in I$ such that $\langle r_1 \rangle \neq I$, then $r_2 \in I \setminus \{r_1\}$, etc. Then $\langle r_1 \rangle \subset \langle r_1, r_2 \rangle \subset \langle r_1, r_2, r_3 \rangle, \ldots$. ∎

Theorem 2.74 (Hilbert's basis theorem)
If R is Noetherian, then $R[X]$ is Noetherian.

Proof. Let $J \triangleleft R[X]$, and let $f_1 \in J$ have minimal degree. If $J = \langle f_1 \rangle$, we're done, so suppose not, and choose $f_2 \in J \setminus \langle f_1 \rangle$ of minimal degree. Continue in this fashion to produce some sequence of f_i's. Let a_i be the leading coefficient of f_i. The chain of ideals $\langle a_1 \rangle \subseteq \langle a_1, a_2 \rangle \subseteq \langle a_1, a_2, a_3 \rangle$ stabilises, since R is Noetherian, so that $\langle a_1, a_2, \ldots \rangle = \langle a_1, \ldots, a_m \rangle$ for some m. Let b_i be such that

$$a_{m+1} = \sum_{i=1}^{m} a_i b_i$$

Now, consider the polynomial

$$g = \sum_{i=1}^{m} b_i f_i x^{\deg f_{m+1} - \deg f_i}$$

which we can do since $\deg f_{m+1} \geq \deg f_i$ as we took the minimal degree at each step. Then g has the same degree as f_{m+1} and the same leading coefficient. So let us consider $f_{m+1} - g$ – it has smaller degree than f_{m+1}, but is not in $\langle f_1, \ldots, f_m \rangle$. This contradicts the minimality of the degree of f_{m+1}. Therefore, the process of choosing these f_i's terminates, so $R[X]$ is finitely generated as required. ∎

Proposition 2.75
Let R be Noetherian and $I \triangleleft R$. Then R/I is Noetherian.

Proof. Let $J \triangleleft R/I$, and $J' \triangleleft R$ be corresponding. If $r_1, \ldots, r_m$ generate J', then, since J is the image of J' under $R \to R/I$, $r_1 + I, \ldots, r_m + I$ generate J. ∎

> **Remark.** It's important to note that subrings of Noetherian rings need not be Noetherian. For example, take $R = \mathbb{C}[x_1, x_2, \ldots]$, and F be the field of fractions of R. Then $R \leq F$, but F is clearly Noetherian as it is a field.

§3 Modules

Let R be a ring, and $(M, +, 0)$ be an abelian group.

Definition 3.1 (Module)**.** A quadruple $(M, \dot{)}$ is an R-module if the operation

$$\begin{aligned} \cdot : R \times M &\to M \\ (r, m) &\mapsto r \cdot m \end{aligned}$$

satisfies

- $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$;
- $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$;
- $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$;
- $1 \cdot m = m$.

Example

- If $R = k$ a field, then an R-module M is the same thing as a k-vector space.
- Any abelian group G is a $\mathbb{Z}$-module by the rule $n \cdot g = \underbrace{g + \cdots + g}_{n \text{ times}}$ if $n \in \mathbb{Z}_0$ and $n \cdot g = \underbrace{(-g) + \cdots + (-g)}_{-n \text{ times}}$ otherwise.
- For any ring, $R^m = R \times \cdots \times R$ is an R-module, since there is a natural notion of scaling by elements of R. *Note that R^m is also a ring in its own right (with pointwise multiplication), but we don't care about that.*
- Let $I \triangleleft R$ be an ideal, then I is an R-module with

$$\begin{aligned} R \times I &\to I \\ (r, x) &\mapsto rx \end{aligned}$$

Similarly, R/I is also an R-module by the obvious operation.

Example

Let V be a k-vector space, and pick $\alpha \in \mathrm{End}(V)$. We can make V into a $k[x]$-module as follows:

$$\begin{aligned} k[x] \times V &\to V \\ (f(x), v) &\mapsto (f(\alpha)(v)) \end{aligned}$$

In other words, $(x, v) \mapsto \alpha(v)$, and the rest comes out of linearity. Note that the structure of the resulting module depends very much on the choice of α.

Example

Another example of a module: take a ring homomorphism $\varphi : R \to S$. This makes S an R-module with

$$\begin{aligned} R \times S &\to S \\ (r, s) &\mapsto \varphi(r)S \end{aligned}$$

For example, we have a homomorphism from $\mathbb{C}[x] \to \mathbb{C}[x]/x^2$ in the obvious way – this makes $\mathbb{C}[x]/x^2$ a $\mathbb{C}[x]$-module. Similar ideas would work for, say $\mathbb{C}[xy]$ and $\mathbb{C}[x]$.

Definition 3.2 (Submodules)**.** Let M be an R-mdoule. A subgroup N of M is called a *submodule* if, for every $n \in N$ and $r \in R$, $r \cdot n$ also lies in N. We write $N \leq M$.

Remark. If R is a field, then an R-submodule is the same as a vector subspace.

Example

Take $R = M = \mathbb{Q}$ and $N = \mathbb{Z} \subseteq \mathbb{Q}$, but this is not a $\mathbb{Q}$-submodule.

Definition 3.3 (Quotient modules)**.** Let $N \leq M$ be an R-submodule. The *quotient R-module* is the abelian group M/N with R-module structure:

$$\begin{aligned} R \times M/N &\to M/N \\ (r, m + N) &\mapsto (rm) + N \end{aligned}$$

Definition 3.4 (Module homomorphisms)**.** Let M and N be R-modules. An *R-module homomorphism* is a group homomorphism $\varphi : M \to N$ such that $\varphi(rm) = r\varphi(m)$ for all $r \in R$, $m \in M$. A bijective R-module homomorphism is called an R-module isomorphism.

Theorem 3.5 (First isomorphism theorem for modules)

Let $\varphi : M \to N$ be a homomorphism of R-modules. Then $\ker\varphi = \{m \in M : \varphi(m) = 0\}$ is an R-submodule of M. Similarly, $\operatorname{im}\varphi = \{\varphi(m) : m \in M\}$ is an R-submodule of N. There is an isomorphism $M/\ker\varphi \cong \operatorname{im}\varphi$ sending $(m + \ker\varphi)$ to $\varphi(m)$.

Theorem 3.6 (Second isomorphism theorem for modules)

Let K, $L \leq M$, then $K + L = \{k + l : k \in K, l \in K\}$ is a submodule of M. There is an isomorphism

$$(K + L)/K \cong L/(K \cap L)$$

Remark. We are using additive notation now, since these are all abelian.

Theorem 3.7 (Third isomorphism theorem for modules)
Let $N \leq L \leq M$ be R-submodules of M. Then

$$M/L \cong (M/N)/(L/N)$$

There is also a correspondence, for a submodule $N \leq M$, between submodules of M/N and submodules of M containing N.

Definition 3.8 (Annihilators)**.** Let M be an R-module. For $m \in M$, its *annihilator* is

$$\operatorname{Ann}(m) = \{r \in R : r \cdot m = 0\}$$

More generally, for $S \subseteq M$, $\operatorname{Ann}(S) = \{r \in R : r \cdot s = 0 \ \forall s \in S\}$. Furthermore,

$$\operatorname{Ann}(M) = \bigcap_{m \in M} \operatorname{Ann}(m)$$

Remark. This is saying that we take a non-zero vector, scale it by a non-zero constant, and it ends up being zero! So in a vector space, all annihilators are empty.

Remark. The annihilator $\operatorname{Ann}(S)$ is an ideal in R.

Example
For example, take $R = \mathbb{Z}$, $M = \mathbb{Z}/n\mathbb{Z}$ with the usual scaling. This has an interesting annihilator – the multiples of n.

Definition 3.9 (Submodules generated by an element)**.** Given an element $m \in M$, the *submodule generated by* M is

$$Rm = \{r \cdot m : r \in R\} \leq M$$

Proposition 3.10
For $m \in M$, there is an isomorphism

$$R/\operatorname{Ann}(m) \cong Rm$$

Proof. Consider the function $R \to M : r \mapsto rm$. This is an R-module homomorphism. Now apply the first isomorphism theorem. ■

Definition 3.11 (Finitely generated modules)**.** An R-module M is finitely generated if there exist $m_1, \ldots, m_k \in M$ such that

$$M = Rm_1 + \cdots + Rm_k = \{r_1 m_1 + \cdots + r_k m_k : r_i \in R\}$$

Remark. For a vector space, this is equivalent to the notion of having finite dimension.

Lemma 3.12

An R-module M is finitely generated if and only if there exists a surjective R-module homomorphism $R^k \to M$ for some k (or, equivalently, if M is a quotient of R^k).

Remark. Modules R^k are sometimes called *free* modules.

Proof. If M is finitely generated, then $M = Rm_1 + \cdots + Rm_k$. Set

$$\begin{aligned} \varphi : R^k &\to M \\ (r_1, \ldots, r_k) &\mapsto r_1 m_1 + \cdots + r_k m_k \end{aligned}$$

which is surjective by hypothesis. Conversely, given a surjective $R^k \to M$, let $e_i = (0, \ldots, 1, \ldots, 0) \in R^k$, where 1 is in the i^{th} position. Define $m_i = \varphi(e_i)$ – it is straightforward to check that $M = Rm_1 + \cdots + Rm_k$. ■

Corollary 3.13

Let M be finitely generated. If $N \leq M$ is a submodule, then M/N is also finitely generated.

Proof. Take the surjection $R^k \to M$, and compose this with the surjective quotient map $M \to M/N$. ■

Remark. Submodules need not have this property. Take $R = \mathbb{C}[x_1, \ldots]$ and $M = R$, which is clearly finitely generated. Take $I \leq M$ to be $\langle x_1, \ldots \rangle$. This is not finitely generated.